

Report - Version 7: Hyperparameter Optimization

Version 7 extends **Building Agentic AutoML** with automatic comparison of model hyperparameter configurations.

The agent preserves task detection, feature detection, Data Inspection, the two preprocessing strategies, the two feature engineering strategies, the three candidate models, and Smart Validation from Version 6. The new capability is to evaluate, for each model, a baseline configuration and one alternative configuration.

With 2 preprocessing strategies, 2 feature engineering strategies, 3 models, 2 configurations, and 5 folds, the agent compares **24 experiments** and performs **120 fits per dataset**. Metrics and validation remain unchanged so that Version 7 introduces only one new concept: **Hyperparameter Optimization**.

Goal of Version 7

- preserve the components introduced in Versions 1-6;
- add small model-specific search spaces for LightGBM, XGBoost, and CatBoost;
- keep the Version 6 configuration as a baseline and compare it with one alternative configuration;
- evaluate every preprocessing x feature engineering x model x hyperparameter combination across 5 folds;
- store hyperparameters, fold scores, mean, and standard deviation for each experiment;
- return best_preprocessing, best_feature_engineering, best_model, best_params, best_score, and best_std.

Architecture

```
Dataset + Target -> Task/Features -> Data Inspection -> Preprocessing  
Strategies -> Feature Engineering -> Model Candidates -> Hyperparameter  
Optimization -> Smart Validation -> Evaluation -> Best Experiment -> State
```

Task Detection

Task detection remains unchanged: at most 20 unique target values means classification; otherwise the task is treated as regression.

```
def detect_task(df, target): n_unique = df[target].nunique() if n_unique <= 20:  
    return "classification" return "regression"
```

Feature Detection

Feature detection also remains unchanged. On Adult Income, the agent detects 6 numerical features and 8 categorical features.

Data Inspection

The Data Inspector introduced in Version 2 is preserved unchanged. On Adult Income it reports 48,842 rows, 15 columns, two target classes, dtypes, missing values, and cardinality. The main missing values remain `workclass = 2,799`, `occupation = 2,809`, and `native_country = 857`.

Preprocessing

Version 7 preserves the **native** and **impute** strategies. Imputation parameters are learned exclusively from the training fold.

Feature Engineering

The Version 6 layer remains unchanged: **none** keeps the original feature space; **interactions** adds pairwise products between numerical features. Original features are always preserved.

Model Selection

LightGBM, XGBoost, and CatBoost remain the three candidates introduced in Version 4.

Hyperparameter Optimization

This is the new capability introduced in Version 7. Each model compares the Version 6 baseline configuration with one predefined alternative configuration:

- **LightGBM**: `n_estimators=200`, `learning_rate=0.05`, `num_leaves=31`;
- **XGBoost**: `n_estimators=200`, `learning_rate=0.05`, `max_depth=4`;
- **CatBoost**: `iterations=500`, `learning_rate=0.05`, `depth=6`.

Metric Selection

Metric selection remains unchanged: ROC AUC for classification and RMSE for regression.

Smart Validation

Validation from Version 5 remains unchanged: **StratifiedKFold** for classification and **KFold** for regression, both with 5 folds, `shuffle=True`, and `random_state=42`.

Training and Evaluation

Every preprocessing-feature engineering-model-hyperparameter combination is evaluated across 5 folds. With 2 preprocessing strategies, 2 feature engineering strategies, 3 models, and 2 configurations, the agent compares 24 experiments and performs 120 fits. Inside each fold: preprocessing -> feature engineering -> fresh model instance with the configuration being evaluated.

Results - Adult Income

Cfg: B = Version 6 baseline; T = Version 7 alternative configuration.

Preproc.	Feature Eng.	Model	Cfg	Mean ROC AUC	Std
native	none	LightGBM	B	0.929556	0.002508
native	none	LightGBM	T	0.929846	0.002751
native	none	XGBoost	B	0.926851	0.002227
native	none	XGBoost	T	0.926509	0.002680
native	none	CatBoost	B	0.930631	0.002246
native	none	CatBoost	T	0.929910	0.002476
native	interactions	LightGBM	B	0.928578	0.002922
native	interactions	LightGBM	T	0.928841	0.002925
native	interactions	XGBoost	B	0.925359	0.002862
native	interactions	XGBoost	T	0.926751	0.003077
native	interactions	CatBoost	B	0.929786	0.002187
native	interactions	CatBoost	T	0.929101	0.002568
impute	none	LightGBM	B	0.929217	0.002713
impute	none	LightGBM	T	0.929522	0.002606
impute	none	XGBoost	B	0.926435	0.002643
impute	none	XGBoost	T	0.926525	0.002548
impute	none	CatBoost	B	0.930119	0.002428
impute	none	CatBoost	T	0.929409	0.002501
impute	interactions	LightGBM	B	0.928319	0.002660
impute	interactions	LightGBM	T	0.928699	0.002746
impute	interactions	XGBoost	B	0.925207	0.002604
impute	interactions	XGBoost	T	0.926060	0.002873
impute	interactions	CatBoost	B	0.929030	0.002533
impute	interactions	CatBoost	T	0.928672	0.002474

Best experiment: **CatBoost + native + none + baseline**, mean ROC AUC = **0.9306306263**, std = **0.0022463105**.

Agent

- load dataset and target; detect task and feature types; inspect the dataset;
- select the metric and validation strategy;
- build candidate models and hyperparameter search spaces;
- evaluate every preprocessing x feature engineering x model x hyperparameter combination across 5 folds;
- store configuration, fold scores, mean, and standard deviation;
- select the best experiment using the mean score and return the final state.

```
{ "task": "classification", "metric": "roc_auc", "validation": "StratifiedKFold",  
  "n_splits": 5, "best_preprocessing": "native", "best_feature_engineering": "none",  
  "best_model": "CatBoost", "best_params": {}, "best_score": 0.930630626301291,  
  "best_std": 0.002246310473216, "experiments": [...] }
```

Regression Test

The same agent is executed on `simple_regression.csv`. It automatically detects regression, selects KFold, uses RMSE, and evaluates all 24 preprocessing-feature engineering-model-hyperparameter combinations.

Cfg: B = Version 6 baseline; T = Version 7 alternative configuration.

Preproc.	Feature Eng.	Model	Cfg	Mean RMSE	Std
native	none	LightGBM	B	1.843523	0.107316
native	none	LightGBM	T	1.822595	0.094805
native	none	XGBoost	B	2.015096	0.065982
native	none	XGBoost	T	1.777285	0.088175
native	none	CatBoost	B	1.663274	0.103853
native	none	CatBoost	T	1.668889	0.090802
native	interactions	LightGBM	B	1.857629	0.117062
native	interactions	LightGBM	T	1.835923	0.128336
native	interactions	XGBoost	B	2.114515	0.127715
native	interactions	XGBoost	T	1.820967	0.098845
native	interactions	CatBoost	B	1.728558	0.113004
native	interactions	CatBoost	T	1.736163	0.128649
impute	none	LightGBM	B	1.856696	0.120398
impute	none	LightGBM	T	1.836213	0.114397
impute	none	XGBoost	B	2.055328	0.130456
impute	none	XGBoost	T	1.797339	0.088521
impute	none	CatBoost	B	1.628342	0.101940
impute	none	CatBoost	T	1.607846	0.100120
impute	interactions	LightGBM	B	1.893113	0.114140
impute	interactions	LightGBM	T	1.852828	0.098626
impute	interactions	XGBoost	B	2.051424	0.112415
impute	interactions	XGBoost	T	1.816835	0.105434
impute	interactions	CatBoost	B	1.681678	0.112769
impute	interactions	CatBoost	T	1.670252	0.104809

Best experiment: **CatBoost + impute + none + tuned**, mean RMSE = **1.6078463494**, std = **0.1001204396**.

What Version 7 Teaches

Version 7 introduces the idea that hyperparameters should also be treated as an AutoML choice and compared empirically. On the classification benchmark, the CatBoost baseline remains best; on the regression benchmark, the alternative CatBoost configuration improves the best RMSE from 1.628342 to 1.607846.

The evolutionary pattern remains explicit: V1 builds the baseline, V2 observes the data, V3 experiments with preprocessing, V4 compares models, V5 makes validation more robust, V6 compares feature spaces, and V7 compares hyperparameter configurations. The **One Version, One Concept** principle remains intact.

Limitations

- the search space is intentionally small: only two configurations per model;
- no Bayesian optimization, random search, or dedicated early stopping;
- cross-validation is fixed to 5 folds, with no group-aware or time-series-aware logic;
- feature engineering remains limited to pairwise numerical products;
- only three boosting model candidates are compared;
- no planner or multi-agent architecture;
- the state does not yet store a persistent fitted pipeline.

Conclusion

Version 7 adds a first Hyperparameter Optimization layer without changing preprocessing, feature engineering, models, metrics, or validation. The agent can now compare configurations, retain their results, and select the best parameters together with the rest of the pipeline. The version prepares the project for Version 8 and a more senior, autonomous agent.